

```
A=randi(10,2,2);
eig(A)
```

```
ans = 2×1
    0.6477
   12.3523
```

```
B=A;
for i= 1:10
    [Q, R] = qr(B,0);
    B=R*Q;
end
```

1

```
A=randi(10,15,15);
eig(A)
```

```
ans = 15×1 complex
 83.0429 + 0.0000i
-11.5112 + 0.0000i
  0.6872 + 9.5985i
  0.6872 - 9.5985i
-3.3486 + 6.0880i
-3.3486 - 6.0880i
-4.1725 + 2.8407i
-4.1725 - 2.8407i
-2.8965 + 0.0000i
 5.5899 + 4.8080i
      ⋮
```

```
B=A;
for i= 1:10
    [Q, R]=qr(B,0);
    B=R*Q;
end
diag(B)
```

```
ans = 15×1
 83.0429
-10.7589
  1.0424
  2.5346
  5.8635
  2.3269
  2.5347
  1.0996
-3.5977
 8.6195
      ⋮
```

2

```
A=randi(10,7,7);
```

```
rank(A)
```

```
ans =  
7
```

```
eig(A)
```

```
ans = 7×1 complex  
38.8869 + 0.0000i  
2.6329 + 6.0650i  
2.6329 - 6.0650i  
3.8278 + 0.0000i  
-6.3856 + 0.0000i  
-2.2974 + 5.4529i  
-2.2974 - 5.4529i
```

```
B=A;  
for i= 1:10  
    [Q, R]=qr(B,0);  
    B=R*Q;  
end  
diag(B)
```

```
ans = 7×1  
38.8869  
-1.0314  
3.2414  
0.3753  
-1.6338  
-6.6614  
3.8231
```

3

a)

```
A=[1 2 1;2 1 3;5 7 0]
```

```
A = 3×3  
    1    2    1  
    2    1    3  
    5    7    0
```

```
eig(A)
```

```
ans = 3×1  
6.7166  
-0.6607  
-4.0559
```

```
B=A;  
for i= 1:50  
    [Q, R]=qr(B,0);  
    B=R*Q;  
end  
diag(B)
```

```
ans = 3×1
    6.7166
   -4.0559
   -0.6607
```

b)

```
A=[1 -1 0 0;-1 2 -1 0;0 -1 2 -1;0 0 -1 2]
```

```
A = 4×4
    1    -1     0     0
   -1     2    -1     0
    0    -1     2    -1
    0     0    -1     2
```

```
eig(A)
```

```
ans = 4×1
    0.1206
    1.0000
    2.3473
    3.5321
```

```
B=A;
for i= 1:50
[Q, R]=qr(B,0);
B=R*Q;
end
diag(B)
```

```
ans = 4×1
    3.5321
    2.3473
    1.0000
    0.1206
```

c)

```
A=randi(10,50,50);
eig(A)
```

```
ans = 50×1 complex
102 ×
    2.7777 + 0.0000i
   -0.1471 + 0.1205i
   -0.1471 - 0.1205i
   -0.1611 + 0.0556i
   -0.1611 - 0.0556i
    0.0092 + 0.1889i
    0.0092 - 0.1889i
   -0.0388 + 0.1818i
   -0.0388 - 0.1818i
   -0.1034 + 0.1378i
      ⋮
```

```
B=A;
for i= 1:50
```

```

[Q, R]=qr(B,0);
B=R*Q;
end
diag(B)

```

```

ans = 50×1
277.7735
20.0470
-8.4799
1.1842
-10.7455
0.3356
-3.2696
0.0470
9.0744
-0.4262
⋮

```

d)

```

A=[1 0 1 0;-1 1 1 -1;1 0 1 0;-1 1 1 -1]

```

```

A = 4×4
    1     0     1     0
   -1     1     1    -1
    1     0     1     0
   -1     1     1    -1

```

```

eig(A)

```

```

ans = 4×1
2.0000
-0.0000
0.0000
-0.0000

```

```

B=A;
for i= 1:50
    [Q, R]=qr(B,0);
    B=R*Q;
end
diag(B)

```

```

ans = 4×1
2.0000
0.0360
-0.0360
0.0000

```

e)

```

A=randi(10,50,50);
rank(A)

```

```

ans =
50

```

```
eig(A)
```

```
ans = 50×1 complex  
102 ×  
 2.7581 + 0.0000i  
 0.1933 + 0.0842i  
 0.1933 - 0.0842i  
-0.1982 + 0.0512i  
-0.1982 - 0.0512i  
-0.0303 + 0.2023i  
-0.0303 - 0.2023i  
 0.1302 + 0.1517i  
 0.1302 - 0.1517i  
 0.1877 + 0.0000i  
  ⋮
```

```
B=A;  
for i= 1:50  
    [Q, R]=qr(B,0);  
    B=R*Q;  
end  
diag(B)
```

```
ans = 50×1  
275.8143  
12.9270  
18.8500  
-13.3276  
-4.5996  
-9.7300  
13.2975  
2.4408  
-0.1364  
14.1829  
  ⋮
```

4

```
A=randi(9,12,7)*randi(9,7,12);  
rank(A)
```

```
ans =  
7
```

```
eig(A)
```

```
ans = 12×1 complex  
103 ×  
 2.0122 + 0.0000i  
 0.0419 + 0.0466i  
 0.0419 - 0.0466i  
-0.0366 + 0.0000i  
-0.0305 + 0.0000i  
 0.0238 + 0.0000i  
 0.0043 + 0.0000i  
 0.0000 + 0.0000i  
 0.0000 + 0.0000i
```

```
0.0000 - 0.0000i
⋮
```

```
B=A;
for i= 1:10
    [Q, R]=qr(B,0);
    B=R*Q;
end
diag(B)
```

```
ans = 12×1
103 ×
    2.0122
    0.0595
    0.0241
   -0.0291
   -0.0367
    0.0228
    0.0043
    0.0000
    0.0000
   -0.0000
    ⋮
    ⋮
```